Website Signals for Institutional Fraud Screening

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Summary

The home-only crawl can already support a first stats sheet, but six bugs blank out or distort key fields. Fixing them and adding 16 capture changes keeps the crawl at one homepage request per site and answers six fraud questions that today's data cannot. A targeted multi-page crawl of the schools the stats sheet flags then supplies the evidence a reviewer needs.

Level	What runs	Requests per site	What it answers
1. Today	Home-only crawl as already run	1 page, plus SSL and WHOIS lookups	Is the site up? How big is the homepage? Which trackers and site builders does it use?
2. Corrected home-only	Same crawl with spec changes C01–C22	1 page (1–2 fallbacks only when the first fails), plus DNS, SSL and WHOIS lookups	Where the URL really leads, who controls the domain, which schools share an operator, whether the site is maintained
3. Targeted deep dive	Multi-page crawl plus outside records, flagged schools only (C23–C27)	Up to 25 pages, plus sitemap and archive lookups	Whether required disclosures and accreditation claims hold up, and what the site's history is

Two companion files go with this document. `SCRIPT_CHANGE_SPEC.md` lists every crawler change by ID (C01–C30) for implementation. `build_stats_sheet.py` builds the stats sheet from `site_features.csv` and starts using Level 2 fields automatically once they exist.

Level 1: what today's data supports

Today's file supports availability, domain, TLD, tracker and homepage-size stats. Eight fields are missing, zeroed or misleading as they stand; the stats script works around each one and labels it in the sheet.

Signal	Usable today?	What to know	Fix
Site reachable, and why not	Yes	Derived from error text and status code. 403, 429 and 503 often mean the site blocks bots, not that it is dead. Sites that serve only HTTP look dead because the crawler forces HTTPS.	C08, C09
Homepage size (words, links, images)	Yes	Column names differ from multi-page mode, so `model_matrix.csv` is missing most derived features. Use `site_features.csv`.	C01, C06
Tracker IDs and site builders present	Yes	Lists are stored as text and must be parsed.	C05
Trackers or templates shared across schools	Recompute only	The precomputed sharing columns are likely all zero in home-only mode and on resumed runs. The stats script recomputes them.	C04
Free-email contact addresses	No	Hard-coded to 0 in home-only mode.	C01
Copyright-year staleness	No	Parsed but never saved.	C01
Contact email on the school's own domain	No	Not saved in home-only mode.	C01
SSL certificate details	Partial	Empty when the crawl ran with `--insecure`, and every site then reads as a basic certificate.	C02
Domain age and WHOIS for broken sites	No	Erased on failed rows, which are the rows where they matter most.	C03

Signal	Usable today?	What to know	Fix
WHOIS privacy	Weak	Registrars now redact most .com and .org records by default, so the flag mostly tracks the TLD.	C12
Where the homepage redirects	No	Only "ended on HTTPS" is stored, not the final address.	C07

Level 2: what a corrected home-only crawl captures

The corrected crawl still fetches one homepage per site, adding a fallback request only when the first one fails, plus DNS, certificate and domain-registration lookups. It answers six questions that matter for fraud screening.

Fraud question	New signals	Example red flag	Changes
Is the site really gone, or just blocking us?	Failure category, bot-protection detection, HTTP and `www.` fallbacks, whether the domain still resolves	The domain no longer resolves, or its name servers point to a domain-parking service	C08, C09, C10
Where does the URL on record actually lead?	Final address after redirects, off-domain redirect flag, parked, placeholder and default-server page detection	The school's URL lands on a "domain for sale" page or an unrelated business	C07, C11
Who controls the domain?	Registrar, creation, expiry and update dates; privacy type (paid proxy vs. default redaction); certificate owner and the other domains on the same certificate	A domain re-registered last year for a school that has operated for decades; one certificate covering several "different" schools	C12, C13

Fraud question	New signals	Example red flag	Changes
Is this school connected to others?	Shared phone numbers, contact emails, street addresses, certificates, identical pages, near-duplicate text, social profiles, registrant organization and redirect targets, plus working tracker and template clustering	Two schools with different owners list the same admissions phone number	C04, C15, C18, C19, C22
Is the site maintained?	Copyright year, server Last-Modified date, certificate and domain expiry	A copyright footer four or more years old, or a domain that expires within 60 days	C01, C12, C17
Does it present like an institution?	Free-email contacts, share of contact email on the school's own domain, accreditation links (recognized vs. not), page title vs. institution name	The admissions contact is a Gmail address, or the homepage cites an "accrediting" body the Department of Education does not recognize	C01, C14, C15, C16

Two changes protect future work rather than adding signals. The crawl saves each homepage's raw HTML (C20), so new features can be extracted later without visiting the sites again. Checkpoints become resume-safe and mode-aware (C05), so a deep-dive run cannot silently skip sites a home-only run already touched.

The stats script picks up each new column automatically once it appears. For example, off-domain redirects, parked pages, stale copyright, free-email contacts and unrecognized accreditor links all become red flags in the score.

Level 3: a deeper dive into each flagged website

A deep dive should run only on the schools the stats sheet flags: the High tier, members of cross-owner clusters, and sites that redirect or look parked. That keeps the number of requests, and the risk of being blocked, proportional to the evidence. The stats script writes this list as `deep_dive_candidates.csv`, which the crawler can read directly.

What a multi-page crawl adds

Signal	What it shows	Source
Required federal disclosures	Whether the site links a net price calculator, consumer information and an annual security report. Title IV schools must publish all three; the calculator applies only to schools enrolling first-time, full-time undergraduates.	C24
Accreditation claims	Which accreditors the site names or links, recognized or not, ready to compare with the school's accreditor of record	C25, C27
Site inventory	True page count, depth, and whether expected sections exist (admissions, registrar, faculty, catalog, library)	Existing sitemap module
Maintenance cadence	Sitemap update dates and catalog dates	Existing sitemap module
Template cloning	Repeated boilerplate across pages, page-template diversity, and CSS files shared with other schools	Existing multi-page features
Broken internal links	Share of sampled internal links that fail	Existing multi-page features
Image reuse	The same campus photo on unrelated schools' sites, and stock-photo markers	C26

What outside records add

Record	What it shows
Internet Archive (Wayback Machine)	When the site first appeared, how often it was captured, and whether the domain once hosted something else

Record	What it shows
Certificate Transparency logs	When the domain first got a certificate, and which other domains appeared on the same certificates
College Scorecard	Operating status, Heightened Cash Monitoring 2 status, ownership, enrollment and the 6-digit OPEID, joined by UNITID
Department of Education accreditation database (DAPIP)	The accreditor of record, to compare with what the website claims

Manual review checklist for each flagged site

1. Confirm the site's status from a browser and over plain HTTP.
2. Compare the address and phone on the site with the Scorecard record.
3. Check the accreditation claim against DAPIP.
4. Look at the Wayback Machine history for a change of owner or purpose.
5. Open the other members of its cluster side by side.

What the data still cannot tell us

Every signal here is a reason to look closer, not proof of fraud. The stats sheet ranks schools for review; a person still makes the call.

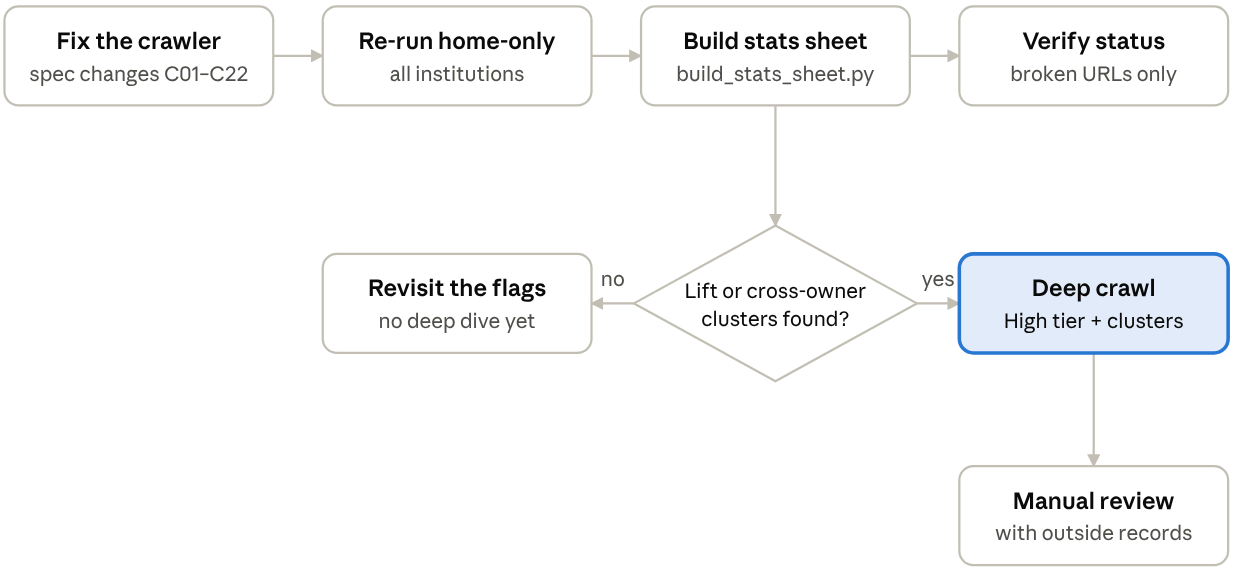
- **Small is not fraudulent.** A legitimate 40-student cosmetology school has a thin homepage. Comparing each school with peers of the same ownership and degree level reduces this, but does not remove it.
- **Sharing is not always suspicious.** Branch campuses, state systems and web vendors legitimately share trackers, templates and registrants. The script excludes pairs under the same 6-digit OPEID and keys shared by more than 25 domains, and it reports registrant clusters without scoring them.
- **Blocked is not dead.** Large universities often block automated visitors. Those sites are counted as alive and get no content-based flags.
- **Registration records are thinning.** Registrars redact most .com and .org WHOIS records by default, and .edu records follow a different format. Missing WHOIS data is common and is not itself a red flag.

- **.edu is a strong but imperfect prior.** New .edu domains require accreditation from a recognized accreditor, but domains registered before October 29, 2001 were grandfathered.
- **Labels are imperfect.** Heightened Cash Monitoring reflects financial or compliance concerns, not fraud, and closed schools' sites go offline for ordinary reasons. The validation tab notes this where it matters.
- **The website is a snapshot.** Sites change, so every finding should carry its crawl date.
- **Claims are not verified by the crawl.** A site naming a recognized accreditor still has to be checked against the accreditor of record.
- **Crawling has limits.** The deep dive respects `robots.txt` and rate limits, and archived homepages need a retention period.

Recommended sequence

Re-run the corrected home-only crawl on every school, then commission the deep dive only if the stats sheet shows real signal, and only for the schools it flags.

A deep dive runs only on flagged schools, and only if the screen shows signal



next-run sequence · 7 steps, 1 decision

The decision gate has two parts. Either the red-flag score separates known-problem schools from the rest (a suggested bar: AUC of 0.65 or higher), or the sheet finds at least one cluster linking schools with different owners. Broken URLs skip the deep crawl and go

to a status check, because a dead site has nothing more to crawl.

Appendix: changes and the fields they add

Full definitions, types and acceptance tests are in `SCRIPT_CHANGE_SPEC.md`.

Change	Priority	What it fixes or captures	Key fields
C01	Fix	Home-only mode writes the same columns as multi-page mode; restores free-email, copyright and contact-email fields	`total_words`, `free_email_hits`, `max_copyright_year`, `mailto_on_domain_ratio`
C02	Fix	Certificate details captured even when verification is off or fails	`ssl_verified`, `ssl_subject_org`, `ssl_not_after`
C03	Fix	Broken sites keep their domain, certificate and WHOIS data	(retention rule)
C04	Fix	Tracker and template sharing work in every mode and on resumed runs	`n_domains_sharing_tracker_id`, `tracker_cluster_id`
C05	Fix	Resume-safe, mode-aware checkpoints; output-path flags	`--checkpoint`, `--output`, `--mode`
C06	Fix	Feature-engineering script recognizes v2 columns	(engineered columns)
C07	Home-only	Where the URL really leads	`final_url`, `redirected_off_domain`
C08	Home-only	Failure category and bot-protection detection	`error_category`, `bot_protection_detected`
C09	Home-only	HTTP and `www.` fallbacks	`http_fallback_used`, `www_fallback_used`

Change	Priority	What it fixes or captures	Key fields
C10	Home-only	DNS resolution and parking name servers	`dns_resolves`, `parking_nameserver_detected`
C11	Home-only	Parked, placeholder and default-server pages	`parked_domain_detected`, `placeholder_page_detected`
C12	Home-only	Registrar dates and privacy type	`whois_creation_date`, `whois_privacy_type`, `domain_days_to_expiry`
C13	Home-only	Certificate owner, lifetime and shared domains	`ssl_san_domains`, `ssl_cert_sha256`
C14	Home-only	Page title and site name	`page_title`, `og_site_name`
C15	Home-only	Contact details for cross-school matching	`phone_numbers`, `contact_emails`, `street_addresses`
C16	Home-only	Accreditation link claims	`unrecognized_accredit_domains`, `accreditors_named`
C17	Home-only	Headers and response time	`has_hsts`, `last_modified_age_days`, `response_time_ms`
C18	Home-only	Page fingerprints	`html_sha256`, `text_simhash64`
C19	Home-only	Social profile links	`social_profile_urls`
C20	Home-only	Raw homepage archive and offline re-extraction	`html_archive_path`, `--from-archive`
C21	Home-only	Run metadata	`crawl_mode`, `script_version`
C22	Home-only	Cross-school matching on all of the above	`n_domains_sharing_<key>`, `operator_cluster_id`
C23	Deep dive	Targeted multi-page runs from the candidate list	(CLI)

Change	Priority	What it fixes or captures	Key fields
C24	Deep dive	Required disclosures and key pages	`has_net_price_calculator_link`, `has_consumer_info_link`, `has_security_report_link`
C25	Deep dive	Accreditation claims across pages	`accreditation_page_url`
C26	Deep dive	Image reuse and stock photos	`image_phashes`, `stock_image_markers`
C27	Deep dive	Wayback, certificate-log and DAPIP enrichment	`wayback_first_capture`, `ct_first_cert_date`, `site_vs_dapip_match`
C28–C30	Docs	README, data dictionary, offline tests, dependencies	—

Sources

- EDUCAUSE: .edu eligibility
- NCES: Net price calculator information center
- AACRAO: Clery Act
- FSA: Department of Education terminates recognition of ACICS (Aug 19, 2022)
- College Scorecard institution data documentation